```
!pip uninstall -y langchain langchain-openai requests
!pip install requests==2.32.4
!pip install langchain langchain-openai langchain-community faiss-cpu tiktoken python-dotenv
```

```
Found existing installation: langchain 1.2.14
Uninstalling langchain-1.2.14:
  Successfully uninstalled langchain-1.2.14
Found existing installation: langchain-openai 1.1.12
Uninstalling langchain-openai-1.1.12:
  Successfully uninstalled langchain-openai-1.1.12
Found existing installation: requests 2.32.4
Uninstalling requests-2.32.4:
  Successfully uninstalled requests-2.32.4
Collecting requests==2.32.4
  Using cached requests-2.32.4-py3-none-any.whl.metadata (4.9 kB)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests==2.32.4) (3.
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests==2.32.4) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests==2.32.4) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests==2.32.4) (2026.2.2
Using cached requests-2.32.4-py3-none-any.whl (64 kB)
Installing collected packages: requests
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is th
langchain-community 0.4.1 requires requests<3.0.0,>=2.32.5, but you have requests 2.32.4 which is incompatible.
Successfully installed requests-2.32.4
Collecting langchain
  Downloading langchain-1.2.15-py3-none-any.whl.metadata (5.8 kB)
Collecting langchain-openai
  Using cached langchain_openai-1.1.12-py3-none-any.whl.metadata (3.1 kB)
Requirement already satisfied: langchain-community in /usr/local/lib/python3.12/dist-packages (0.4.1)
Requirement already satisfied: faiss-cpu in /usr/local/lib/python3.12/dist-packages (1.13.2)
Requirement already satisfied: tiktoken in /usr/local/lib/python3.12/dist-packages (0.12.0)
Requirement already satisfied: python-dotenv in /usr/local/lib/python3.12/dist-packages (1.2.2)
Requirement already satisfied: langchain-core<2.0.0,>=1.2.10 in /usr/local/lib/python3.12/dist-packages (from langchain) (1.2.
Collecting langgraph<1.2.0,>=1.1.5 (from langchain)
  Downloading langgraph-1.1.6-py3-none-any.whl.metadata (8.0 kB)
Requirement already satisfied: pydantic<3.0.0,>=2.7.4 in /usr/local/lib/python3.12/dist-packages (from langchain) (2.12.3)
Requirement already satisfied: openai<3.0.0,>=2.26.0 in /usr/local/lib/python3.12/dist-packages (from langchain-openai) (2.30.
Requirement already satisfied: langchain-classic<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from langchain-commu
Requirement already satisfied: SQLAlchemy<3.0.0,>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from langchain-community)
Collecting requests<3.0.0,>=2.32.5 (from langchain-community)
  Using cached requests-2.33.1-py3-none-any.whl.metadata (4.8 kB)
Requirement already satisfied: PyYAML<7.0.0,>=5.3.0 in /usr/local/lib/python3.12/dist-packages (from langchain-community) (6.0
Requirement already satisfied: aiohttp<4.0.0,>=3.8.3 in /usr/local/lib/python3.12/dist-packages (from langchain-community) (3.
Requirement already satisfied: tenacity!=8.4.0,<10.0.0,>=8.1.0 in /usr/local/lib/python3.12/dist-packages (from langchain-commu
Requirement already satisfied: dataclasses-json<0.7.0,>=0.6.7 in /usr/local/lib/python3.12/dist-packages (from langchain-commun
Requirement already satisfied: pydantic-settings<3.0.0,>=2.10.1 in /usr/local/lib/python3.12/dist-packages (from langchain-comm
Requirement already satisfied: langsmith<1.0.0,>=0.1.125 in /usr/local/lib/python3.12/dist-packages (from langchain-community)
Requirement already satisfied: httpx-sse<1.0.0,>=0.4.0 in /usr/local/lib/python3.12/dist-packages (from langchain-community) (0
Requirement already satisfied: numpy>=1.26.2 in /usr/local/lib/python3.12/dist-packages (from langchain-community) (2.0.2)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from faiss-cpu) (26.0)
Requirement already satisfied: regex>=2022.1.18 in /usr/local/lib/python3.12/dist-packages (from tiktoken) (2025.11.3)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3-
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langcha
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchain
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langch
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->lang
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langcha
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langch
Requirement already satisfied: marshmallow<4.0.0,>=3.18.0 in /usr/local/lib/python3.12/dist-packages (from dataclasses-json<0.7
Requirement already satisfied: typing-inspect<1,>=0.4.0 in /usr/local/lib/python3.12/dist-packages (from dataclasses-json<0.7.0
Requirement already satisfied: langchain-text-splitters<2.0.0,>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from langcha
```

```
from langchain_community.llms import HuggingFacePipeline
from transformers import pipeline

# Load free model
pipe = pipeline("text-generation", model="gpt2")

llm = HuggingFacePipeline(pipeline=pipe)

from langchain_core.prompts import PromptTemplate

prompt = PromptTemplate.from_template(
    "Explain {topic} in simple terms"
)

chain = prompt | llm
```

```
response = chain.invoke({"topic": "Artificial Intelligence"})
print(response)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to en
config.json: 100%  665/665 [00:00<00:00, 12.0kB/s]
model.safetensors: 100%  548M/548M [00:07<00:00, 170MB/s]
Loading weights: 100%  148/148 [00:00<00:00, 282.50it/s, Materializing param=transformer.wte.weight]
GPT2LMHeadModel LOAD REPORT from: gpt2
Key                  | Status     |  |
---------------------+------------+--+-
h.{0...11}.attn.bias | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
generation_config.json: 100%  124/124 [00:00<00:00, 2.07kB/s]
tokenizer_config.json: 100%  26.0/26.0 [00:00<00:00, 400B/s]
vocab.json:  1.04M/? [00:00<00:00, 10.8MB/s]
merges.txt:  456k/? [00:00<00:00, 3.45MB/s]
tokenizer.json:  1.36M/? [00:00<00:00, 15.1MB/s]
/tmp/ipykernel_15525/3606093153.py:7: LangChainDeprecationWarning: The class `HuggingFacePipeline` was deprecated in LangChain 0
  llm = HuggingFacePipeline(pipeline=pipe)
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
Both `max_new_tokens` (=256) and `max_length`(=50) seem to have been set. `max_new_tokens` will take precedence. Please refer to
Explain Artificial Intelligence in simple terms, but with deep learning, if we're going to be able to do deep learning with real

You can find out about this research here.

And that's just my personal experience and that's what I hope gets more coverage. I think there's a lot of work that goes into A

My point is that there are things that you can't do from any other point of view. When you're working on that database, you can'

So, there's a lot of work that goes into building that database. But I am not going to talk about it here, because I think it
```

```
%pip install -U langchain-huggingface
```

```
Collecting langchain-huggingface
  Downloading langchain_huggingface-1.2.1-py3-none-any.whl.metadata (3.3 kB)
Requirement already satisfied: huggingface-hub<2.0.0,>=0.33.4 in /usr/local/lib/python3.12/dist-packages (from langchain-hugging
Requirement already satisfied: langchain-core<2.0.0,>=1.2.11 in /usr/local/lib/python3.12/dist-packages (from langchain-huggingf
Requirement already satisfied: tokenizers<1.0.0,>=0.19.1 in /usr/local/lib/python3.12/dist-packages (from langchain-huggingface)
Requirement already satisfied: filelock>=3.10.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0.0,>=0.33.4-
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0.0,>=0.33.4-
Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0.0,>=0.3
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0.0,>=0.33.4-
Requirement already satisfied: packaging>=20.9 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0.0,>=0.33.4->
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0.0,>=0.33.4->lang
Requirement already satisfied: tqdm>=4.42.1 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0.0,>=0.33.4->lan
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0.0,>=0.33.4->langchain-
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0.0,>
Requirement already satisfied: jsonpatch<2.0.0,>=1.33.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=
Requirement already satisfied: langsmith<1.0.0,>=0.3.45 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=
Requirement already satisfied: pydantic<3.0.0,>=2.7.4 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.
Requirement already satisfied: tenacity!=8.4.0,<10.0.0,>=8.1.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2
Requirement already satisfied: uuid-utils<1.0,>=0.12.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0.0,>=
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0.0,
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0.0,>=0
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggi
Requirement already satisfied: jsonpointer>=1.9 in /usr/local/lib/python3.12/dist-packages (from jsonpatch<2.0.0,>=1.33.0->langc
Requirement already satisfied: orjson>=3.9.14 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.3.45->langcha
Requirement already satisfied: requests-toolbelt>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.3.4
Requirement already satisfied: requests>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.3.45->langch
Requirement already satisfied: xxhash>=3.0.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.3.45->langchai
Requirement already satisfied: zstandard>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.3.45->lang
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.7.4->l
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.7.4->la
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.7.4-
```

```
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0.0,>=0.33
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0.0,
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0.0,>=0.33
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0.
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.0.0->langsm
Requirement already satisfied: urllib3<3,>=1.26 in /usr/local/lib/python3.12/dist-packages (from requests>=2.0.0->langsmith<1.0.
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggi
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->hug
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0->
Downloading langchain_huggingface-1.2.1-py3-none-any.whl (30 kB)
Installing collected packages: langchain-huggingface
Successfully installed langchain-huggingface-1.2.1
```

```
from langchain_huggingface import HuggingFacePipeline
```

Clean LLM Setup

```
from langchain_huggingface import HuggingFacePipeline
from transformers import pipeline

pipe = pipeline("text-generation", model="gpt2")

llm = HuggingFacePipeline(pipeline=pipe)
```

```
Loading weights: 100%                                    148/148 [00:00<00:00, 256.94it/s, Materializing param=transformer.wte.weight]
GPT2LMHeadModel LOAD REPORT from: gpt2
Key                  | Status     |  |
---------------------+------------+--+-
h.{0...11}.attn.bias | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
```

Clean Prompt + Chain

```
pipe = pipeline(
    "text-generation",
    model="gpt2",
    max_new_tokens=100,
    temperature=0.7
)

llm = HuggingFacePipeline(pipeline=pipe)
```

```
Loading weights: 100%                                    148/148 [00:00<00:00, 499.24it/s, Materializing param=transformer.wte.weight]
GPT2LMHeadModel LOAD REPORT from: gpt2
Key                  | Status     |  |
---------------------+------------+--+-
h.{0...11}.attn.bias | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
```

```
# Claen Output (Trim Garbage Text)
def clean_output(text):
    return text.split("\n")[0]

result = chain.invoke({"topic": "Artificial Intelligence"})
print(clean_output(result))
```

```
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
Both `max_new_tokens` (=100) and `max_length`(=50) seem to have been set. `max_new_tokens` will take precedence. Please refer to
Explain Artificial Intelligence in simple and clear points.
```

```
# Memory (Improved Version)
chat_history = []

def chat(user_input):
    chat_history.append(f"User: {user_input}")

    context = "\n".join(chat_history)

    prompt = f"""
```

```
    Conversation:
    {context}

    AI:
    """

    response = llm.invoke(prompt)
    response = clean_output(response)

    chat_history.append(f"AI: {response}")

    return response

print(chat("Hi, I am Murali"))
print(chat("What is my name?"))
```

```
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
Both `max_new_tokens` (=100) and `max_length`(=50) seem to have been set. `max_new_tokens` will take precedence. Please refer to
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
Both `max_new_tokens` (=100) and `max_length`(=50) seem to have been set. `max_new_tokens` will take precedence. Please refer to
```

```
# Tool (Calculator)
def calculator(query):
    try:
        return str(eval(query))
    except:
        return "Invalid expression"

print(calculator("45*2+10"))
```

```
100
```

```
# Agent (Better Logic)
def agent(query):
    if any(char.isdigit() for char in query):
        return calculator(query)
    else:
        return clean_output(llm.invoke(query))

print(agent("What is 25 * 4 + 10?"))
print(agent("Explain LangChain"))
```

```
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
Both `max_new_tokens` (=100) and `max_length`(=50) seem to have been set. `max_new_tokens` will take precedence. Please refer to
Invalid expression
Explain LangChain
```

```
# Document Loader
from langchain_community.document_loaders import TextLoader

with open("knowledge.txt", "w") as f:
    f.write("""
    LangChain is a framework for building applications using Large Language Models.
    It supports chains, memory, agents, and retrieval-based systems.
    """)

loader = TextLoader("knowledge.txt")
documents = loader.load()
```

```
# Text Splitting
from langchain_text_splitters import CharacterTextSplitter

splitter = CharacterTextSplitter(chunk_size=100, chunk_overlap=20)
docs = splitter.split_documents(documents)
```

```
# Vector Store
from langchain_community.vectorstores import FAISS
from langchain_community.embeddings import HuggingFaceEmbeddings

embeddings = HuggingFaceEmbeddings()

vectorstore = FAISS.from_documents(docs, embeddings)
```

```
/tmp/ipykernel_15525/729768178.py:4: LangChainDeprecationWarning: Default values for HuggingFaceEmbeddings.model_name were depre
  embeddings = HuggingFaceEmbeddings()
Loading weights: 100%|| 199/199 [00:00<00:00, 484.35it/s, Materializing param=pooler.dense.weight]
MPNetModel LOAD REPORT from: sentence-transformers/all-mpnet-base-v2
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
```

```
# RAG System
def rag(query):
    docs = vectorstore.similarity_search(query)
    context = docs[0].page_content

    prompt = f"""
    Answer using context:

    {context}

    Question: {query}
    """

    return clean_output(llm.invoke(prompt))

print(rag("What is LangChain?"))
```

```
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
Both `max_new_tokens` (=100) and `max_length`(=50) seem to have been set. `max_new_tokens` will take precedence. Please refer to
```

Start coding or generate with AI.